

Letter of Intent & Research Statement

Applicant: Rui Zhu

Proposed Program: Exploratory Research (Supplementing a thesis) in Mathematics

Affiliation: Guangdong Technion – Israel Institute of Technology (GTIIT)

Proposed Supervisors: Dr. Marcelo Ciappina & Dr. Gilad Gour

Dear Members of the Graduate Committee,

I am writing to express my interest in pursuing Exploratory Research under the supervision of Professor Marcelo Ciappina & Professor Gilad Gour at GTIIT. My research interest is the uncertainty relations for two-outcome quantum measurements, studied through the lens of majorization theory.

The research question I would like to start from, is that, given a pair of projective qubit measurements, how can we find an elegant description regarding the joint probability distributions of all quantum states? Current classical entropic formulations—such as the Maassen–Uffink bound—capture this through a single scalar inequality, which is clean and effective but inevitably discards information. Majorization can be a more efficient tool specifically for this problem: it compares distributions at the level of their full ordering, and any Schur-convex function (including Shannon, Rényi, and min-entropies) then inherits the bound automatically. The consequent of this exploratory research that I hope to come up with, is to find a state-independent bounding distribution $\omega(M, N)$ such that the direct sum of outcome distributions for any state ρ is majorized by $\omega(M, N)$, and to understand when this bound is tight and which states reach this bound.

What draws me to this problem is that it sits at the intersection of things I find passionately interesting: the convex geometry of quantum state space, the spectral theory of operators, and the question of what “uncertainty” really means at a structural level rather than just as a number. The qubit case is already non-trivial—the achievable set $\wp(M, N)$ is convex and its boundary is traced by pure states, but characterizing its majorization expression requires careful investigation. Extending this to general two-outcome POVMs via Naimark dilation adds another layer of structure and becomes a more forward-looking goal I am eager to explore.

Moreover, it is worth noting that the structural nature of this majorization-based framework for quantum mechanics can also provide a great paradigm for other subjects. By prioritizing distributional comparisons over scalar statistical summaries, the techniques developed here may eventually provide a valuable lens for investigating ordering relations in other information-theoretic contexts, including the analysis of attention mechanisms in machine learning models, thereby positioning this research at the intersection of quantum theory and complex information processing.

I have a strong background in mathematics, and I have spent the past year building familiarity with quantum information theory, linear algebra, and convex optimizations. I am comfortable working with the mathematical expressions and I am delighted to spend time on the technical details. The twelve-month timeline has been evaluated as a realistic and rational planning—the qubit case as a first milestone, then the POVM extension, with a preprint targeting Physical Review A or Quantum as a concrete deliverable.

This would be the statement of my application. By careful feasibility estimation and timeline management, this project is considered both well-scoped and scientifically meaningful, and I am motivated to put effort in this research interest.

Sincerely,

A handwritten signature in cursive script that reads "Rui Zhu".

Rui Zhu

M.S. in Computer Science, UT Austin